

Future Climate Interventions Simulation for Food Production: Pear Quality

Project: Multivariate and Hierarchical Data (3565)

Project Discovering Associations

2025-2026

1st year Master of Statistics

Hasselt University

Group members:

Luu Trong Nghia (2502924)

Brohi Abdullah Ali (2501314)

Hossain Tanjim (2469700)

Andryan David (2502985)

Tahri Mohammed (2500355)

Submission Date: 24 May, 2026

Lecturers:

Prof. Dr. Geert Molenberghs

Prof. Dr. Olivier Thas

Prof. Dr. Geert Verbeke

Prof. Dr. Yannick Vandendijck

Prof. Dr. Steven Abrams

Prof. Dr. Johan Verbeeck

Roel Jude Bagaforo

Abstract

Climate change had been an issue to agricultural sector, particularly for fruit crops such as pears that were sensitive to the environmental condition. In this study, we investigated the effect of four future climate scenarios on pear quality using Ecotron facilities in Belgium and France. There were three main objectives: assess pear quality, analyzing climate conditions to the size of the pears over time, and exploring soil conditions. To address those objectives, we used Linear Mixed Model (LMM) and Generalized Linear Mixed Model (GLMM) with Bernoulli assumption. Additionally, Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) were used to explore the soil conditions across climate scenarios. The results showed that Doyenné variety had a lower quality index by 6.35 and slower growth rate by 0.05 cm every week, compared to Conference variety. Additionally, the second climate scenario was the only intervention that was estimated from all statistical models to have a significant positive effect. It improve the quality index by 9.99, the odds of good quality was 1.67 times and estimated to have a faster growth rate of 0.0052 per week. The soil analysis did not reveal strong separation between climate scenarios in PCA, though LDA showed some discriminatory power driven primarily by soil physical characteristics. Overall, these findings suggest that scenario 2 to be the best climate scenario in the future.

Key Words: Ecotron; pears; mixed model; climate; agriculture

1 Introduction

Climate change represented a threat to agricultural sector on a global scale. In Belgium, to ensure long-term food production sustainability pears were used as a sample of experiment, since pears were quite sensitive to the environment conditions [1]. The experiment was carried out in Ecotron facilities, which are sophisticated experimental units that allow precise manipulation of environmental variables such as temperature, CO₂ levels, humidity, and soil conditions [2]. These facilities make them ideal for simulating plausible future climates. A total of four climate scenarios, representing possible conditions in 2070, were applied across the Ecotrons. Each of these scenarios reflected a different type of climate intervention: no major intervention to strategies focused on CO₂ removal, sustainable energy, and transportation reform. Two common pear varieties in Europe, Conference and Doyenné du Comice were planted in Belgium and France under those interventions [3]. Additionally, since the main variable of interest was the climate scenario, the pear species was identified as a potential confounder that might help explaining the pear quality and size. The aim of this study was to address which climate intervention scenario is the best for pear quality in the future.

There were three endpoints. The first was to assess which is the best scenario for pear quality in the future in continue and binary outcome. The second was to analyze of the effect of the climate conditions on the “size” of the pears over time. The last endpoint was to explore the soil condition features with a multivariate method to gain insights into the effect of climate on the pear production.

2 Methods and Materials

2.1 Data Source

To address the study objectives, three distinct datasets were used. The first dataset had a total of 1942 pears. This dataset followed a hierarchical structure comprising 16 Ecotrons and 96 trees. There were 8 Ecotrons located in Belgium, and the rests were in France. Each Ecotron had 6 trees, three trees each species. Moreover, each Ecotron was assigned to specifically one of the four climate scenarios and no changed of scenario in the middle of study period. This means each climate was replicated two times for each location. To investigate the additional aim on the first study aim, the quality metric was categorized: pears with a score of 55 or higher were classified as “Good” (1), while those scoring below 55 were classified as “Bad” (0).

The second dataset was a longitudinal data with 192 pears measured repeatedly over 20 weeks (from week 5 to 24). This experimental had a different data structure from the first dataset where four climate scenarios are uniquely assigned to four specific ecotrons (one climate per unit) and here we did not have location variable. Additionally, each ecotron contained two trees, representing the two species, *Conference* and *Doyenné du Comice*.

The longitudinal dataset had incompleteness measurement of pears called dropout. Figure S2 shown some pears of each variety dropping-out from the study started at week 7. At the end of the study, only 94 Conference and 79 Doyenné left in the study. This meant around 90% pears left in total from 192 pears.

The third dataset comprised multivariate soil data, where each ecotron was characterized by a diverse type of soil The study included 96 trees with 8 units in Belgium and 8 in France. Every ecotron contained six trees in total and three each pear species.

2.2 Hypotheses

The null hypothesis was no significant difference between all the climate scenarios in terms of pear quality. While the alternative is at least one future climate scenario results in a higher mean pear quality index compared to the control scenario 1. Let β_i denoted the

effect of climate scenario i relative to scenario 1,

$$\begin{cases} H_0 : \beta_2 = \beta_3 = \beta_4 = 0, \\ H_1 : \beta_i > 0, \quad \exists i \in \{2, 3, 4\}. \end{cases}$$

Moreover, the significance level was set to $\alpha = 0.05$, but the Bonferroni correction was also applied to account the multiple comparisons of climate scenario 1 with other scenarios, such that the adjusted significance level was $\alpha = \frac{0.05}{3} = 0.0167$.

2.3 Sample Size Calculation

From the pilot data, we had four Ecotrons in two different location, Belgium and France, two species of pears, with a total of 30 Conference and 30 Doyenné. The simulation was done on ten thousands iterations by applying specified parameters:

- Effect size (δ): 10, 15, and 20;
- Statistical power in a range of 0.80 to 0.90.

2.4 Statistical Model

We considered four different models to address the study objectives. Let Y_{ijk} be the quality of the i -th pear for the j -th tree from the k -th Ecotron. Hence, the Linear Mixed Model (LMM) was written as

$$Y_{ijk} = \beta_0 + b_{0j} + \beta_1 \text{Species}_{ijk} + \sum_{m=2}^4 \beta_m \text{Climate}_{mijk} + \beta_5 \text{Location}_{ijk} + \varepsilon_{ijk}, \quad (1)$$

where $i = 1, \dots, n$ number of pears, and β_0 represented the average pear quality for the reference group, *Conference* variety of scenario 1 in Belgium. Additionally, we assumed random intercept on tree _{j} $b_{0j} \sim N(0, \sigma_{tree}^2)$ and residual error $\varepsilon_{ijk} \sim N(0, \sigma_{res}^2)$ [4].

Furthermore, we used a Generalized Mixed Model (GLMM) for estimating “good” pear quality with assumption of $Y_{ijk}|b_{0j} \sim \text{Bernoulli}(\pi_{ijk})$,

$$\text{logit}(\pi_{ijk}) = \beta_0 + b_{0j} + \beta_1 \text{Species}_{ijk} + \sum_{m=2}^4 \beta_m \text{Climate}_{mijk} + \beta_5 \text{Location}_{ijk}, \quad (2)$$

where $b_{0j} \sim N(0, \sigma_{tree}^2)$ and β_0 represented the reference group of Conference variety planted in Belgium under climate scenario 1 [5].

For fitting the longitudinal data, let Y_{ijt} be the size in centimeters (cm) of the i -th pear from the j -th tree at the t -th week. We write the LMM for Gaussian longitudinal data as

$$\begin{aligned} Y_{ijt} = & \beta_0 + b_{0j} + b_{0i(j)} + \beta_1 \text{Species}_{ijt} + \sum_{m=1}^3 \beta_{2m} \text{Climate}_{im} + (\beta_3 + b_{1i(j)}) \text{Week}_{ijt} \\ & + \sum_{l=1}^3 \beta_{4l} \text{Climate}_{ijl} \cdot \text{Week}_{ijt} + \beta_5 \text{Species}_{ijt} \cdot \text{Week}_{ijt} + \varepsilon_{ijt}, \end{aligned} \quad (3)$$

where Conference variety and climate scenario 1 as the reference group [5], with

- $b_{0j} \sim N(0, \sigma_{tree}^2)$ was the deviation of the j -th tree from the overall intercept;
- $b_{0i(j)}$ was the deviation of the i -th pear nested in the j -th tree from the overall intercept;
- $b_{1i(j)}$ expressed how much the slope of the i -th pear nested in the j -th tree deviated from the average slope in the climate (ecotron) to which the pear belongs;
- $\mathbf{b}_i = (b_{0i(j)}, b_{1i(j)})' \sim N(0, D)$;
- $\varepsilon_{ijt} \sim N(0, \sigma_{res}^2)$.

The matrix D was a 2×2 covariance matrix of the pear random effect,

$$D = \begin{bmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{bmatrix},$$

and the correlation between intercept and slope defined as

$$\text{corr}(b_{0i(j)}, b_{1i(j)}) = \frac{d_{12}}{\sqrt{d_{11}}\sqrt{d_{22}}}. \quad (4)$$

For the likelihood based model, model selection was performed using the Likelihood ratio test (LRT). For non-likelihood based model, model selection for the GEE was performed using the quasi-likelihood under independence model criterion (QIC) for working correlation structure selection [6].

2.5 Multivariate Methods

To explore the structure of soil condition features and assess their potential relationship with climate scenarios and pear production, Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) was applied [7, 8]. PCA was used as an unsupervised dimensionality reduction technique to investigate the underlying variability of the 17 soil

features without incorporating climate group information. All variables were standardized prior to analysis to ensure comparability across different measurement scales. PCA transforms the original correlated variables into a set of orthogonal principal components that capture the maximum variance in the data [8]. A scree plot was used to evaluate the proportion of variance explained by each component and to assess the effective dimensionality of the dataset, while a biplot of the first two components was used to visualize the distribution of observations and the contribution of soil variables.

Subsequently, LDA was performed as a supervised method to evaluate whether soil characteristics, together with pear production (measured as the number of pears), can discriminate between predefined climate scenarios. In this analysis, the climate scenario was treated as the grouping variable, while soil features and pear counts were included as predictors, allowing both direct and indirect effects of climate (via soil conditions) to be captured. LDA identifies linear combinations of predictors that maximize the separation between groups relative to within-group variability [8]. The results were assessed through eigenvalue decomposition to quantify the proportion of between-group variance explained by each discriminant function, through a scatter plot of the first two discriminants and through a visualization of the LD1 loadings to assess the contribution

3 Result

3.1 Sample Size

Based on the statistical power analysis, the minimum effect size (δ) of 10 and ICC of 0.20 shown in Figure S1 and Table 1 resulted sample size of 540 pears, including for potential unforeseen circumstances. The study was involved 16 ecotrons located in Belgium and France. Each ecotron contained six trees, providing an equal distribution of the two species under study: three Conference and three Doyenné trees per unit. The experimental framework used 96 trees and a total sample of 2,160 pears. However, in the data there were only 1942 pears.

Table 1: Sample size calculation results with statistical power ranged from around 80% to 90%.

Sample size	400	450	500	550
Power	79.41%	84.27%	88.56%	90.89%

3.2 Exploratory Data Analysis

The analysis included three different datasets, covering the no time dependent (1), longitudinal (2), and multivariate case (3). Each dataset had different size and endpoints that explained in Section 2. Before proceeding to the modeling stage, each dataset was explored.

Table S1 described the mean, quartiles, and interquartile range (IQR) of the quality index for climate scenario, location, and species. The IQR across majority cases ranged from around $Q_1 \approx 30$ to $Q_3 \approx 75$, indicating a similar variability of pear quality regardless of the experimental conditions. This was supported by Figure 1a which illustrated similar variability across all climates and locations, with quality index ranging from 0 to 100 and an overall mean of 56.09.

Pears planted in Belgium had a slightly higher average quality (58.11) by around 6 units difference to those in France (53.22). At the species level in Belgium, Conference averaged 61.15 against 55.95 for Doyenné, while in France the same structure was observed with averages of 54.52 and 49.93 respectively. Notably, scenario 2 showed the highest mean

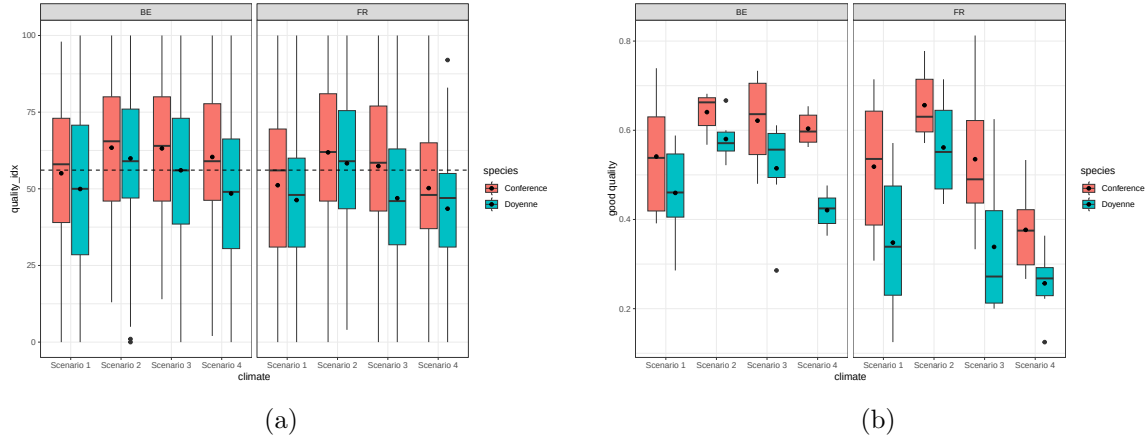


Figure 1: Distribution of pears quality index in every location and climate scenario. The dashed-line showed the overall mean for the Gaussian outcome.

quality index with Belgium Conference reaching 63.41 (Q3) and France Conference 61.89. On the other hand, scenario 4 consistently showed slightly lower mean to Scenario 1. Despite the numerical difference, Figure 1a showed no strong visual indication of interaction between any combination of species, climate, and location.

For the binary outcome, we dichotomized the quality index with 55 or more indicated as a good quality. Figure 1b showed the proportion of good quality pears exhibited more visible variability across subgroups, particularly between scenario 4 and location, indicating a potential interaction between these two factors for the binary endpoint.

Figure 2a illustrated that at the first week all climate scenarios started at a similar point. However, as the weeks progressed, the second climate scenario showed a faster average growth compared to other climates. On the other hand, climate 1, 3, and 4 had a similar growth average. Notably, Figure 2b showed that the Conference variety had a higher average size from the starting week than the Doyenné and consistently had a faster average growth over the week.

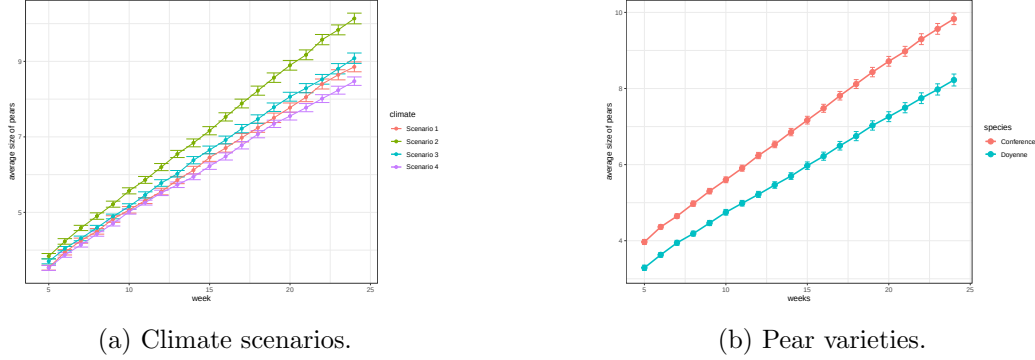
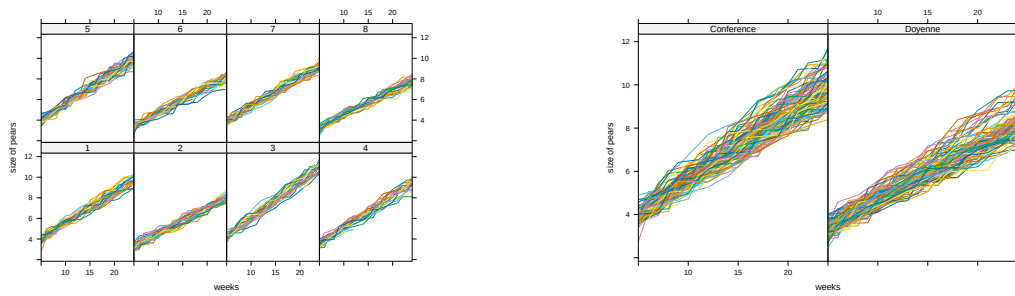


Figure 2: The average size of pears by climate and varieties for 20 weeks study period.

The pears size specific was showed by Figure 3a and 3b. The growth of each pear had a different pace between trees. Pears in tree number 1, 3, and 5 grew relatively faster than those in other trees, indicating a random effect at tree level. Additionally, each pear in every variety shown in Figure 3b had different starting points and different growth rates across the 20 weeks of study period, consequently indicating a randomness each pear nested in trees. Figure 3b also illustrated that Conference variety grew faster than Doyenné.



(a) Growth of each pear size nested in trees. (b) Size of pears specific for each species.

Figure 3: Pears size specific distribution for the entire study period.

3.3 The effect of climate scenario on the pear quality index

Linear Mixed Model (1) was fitted under the compound symmetry (CS), where the estimated $\widehat{ICC} = 0.0084$. The $\widehat{ICC}$ suggested the tree level clustering had small contribution to the variation in pear quality. Furthermore, the fixed effect result of the fitted model was described in Table 2. The table showed the significant effect of independent variables on the pear quality index.

Table 2: Type 3 test of fixed effects

Effect	<i>F</i> -value	<i>Pr</i> > <i>F</i>
Species	29.21	<0.0001
Climate	16.99	<0.0001
Location	16.30	<0.0001

The parameter estimates for the fixed effect were shown in Table 3. The intercept represented the estimated average quality index of 56.09 for Conference variety planted in Belgium under scenario 1. Unlike Conference, Doyenné variety had a significantly lower pear quality index by -6.3519 . Furthermore, pears planted in the Ecotron located in France had a quality index was estimated to be 4.7883 lower compared to Belgium. Table S2 de-

Table 3: Parameter estimates (standard error) for pear quality index outcome. The stars showed whether the parameters were significant or not, across the models.

Parameter	LMM (CS)			GEE (Ind.)			GEE (Exch.)		
Intercept	56.0887 (1.4862)	*		56.0504 (1.5937)	*		56.0704 (1.5951)	*	
Doyenne	-6.3519 (1.1752)	*		-6.3241 (1.1154)	*		-6.3389 (1.1222)	*	
Scenario 2	9.9964 (1.6390)	*		9.8645 (1.4969)	*		9.9304 (1.5002)	*	
Scenario 3	5.5843 (1.7082)	*		5.6666 (1.9522)	*		5.6252 (1.9532)	*	
Scenario 4	0.4953 (1.8087)			0.5565 (1.6599)			0.5264 (1.6586)		
France	-4.7883 (1.1861)	*		-4.7134 (1.1952)	*		-4.7515 (1.1956)	*	
	$\widehat{ICC} = 0.0084$			$\hat{\rho} = 0$			$\hat{\rho} = 0.0038$		
				QIC = 1949.2754			QIC = 1949.3339		

scribed the contrast between climate scenario 1 and other scenarios. Under the Bonferroni correction, the adjusted significance level of 0.0167 for the three pairwise climate scenario comparisons only scenario 2 and 3 showed a significantly higher quality index compared to the scenario 1. Scenario 2 yielded the largest improvement, with an estimated difference of 9.9964 and Scenario 3 with 5.5843. Scenario 4, however, did not differ meaningfully from Scenario 1.

3.4 The effect of climate scenario on good pear quality

The GLMM defined by (2) was fitted with under the compound symmetry (CS) structure. The $\widehat{ICC}$ estimated at 0.0002 indicated that pears within the same tree were less correlated with respect to the binary quality outcome.

Table 4: Type 3 test of variables.

Effect	χ^2	$Pr > \chi^2$
Species	27.72	<0.0001
Climate	14.81	<0.0001
Location	13.31	0.0003

Table 4 illustrated that all three variables had a significant effect on the odds of having good pear quality. Furthermore, the parameter estimates of the fitted model were presented in Table 5. The intercept represented the odds of producing a good quality for the reference group was 1.3536 or 35.36. Unlike the Conference, the odds of producing a good quality pear for Doyenné variety was $e^{-0.4894} \approx 0.6129$ times the reference group. In other words, Doyenné had approximately 39% lower odds of good quality compared to Conference.

Additionally, we observed that location had a significant effect in the model. Holding all other variables constant, pears were planted in France had a lower odds of good quality by 0.6987 times to Belgium. This meant having a good pear quality from France was almost 30% lower than pears planted in Belgium.

Table 5: Parameter estimates (standard error) for binary pear quality index outcome. The stars showed whether the parameters were significant or not, across the models.

Parameter	GLMM (CS)		GEE (Ind.)		GEE (Exch.)	
Intercept	0.3028 (0.1180)	*	0.3028 (0.1327)	*	0.2999 (0.1326)	*
Doyenné	−0.4894 (0.0931)	*	−0.4893 (0.0892)	*	−0.4865 (0.0885)	*
Scenario 2	0.5209 (0.1300)	*	0.5208 (0.1305)	*	0.5151 (0.1301)	*
Scenario 3	0.1723 (0.1353)		0.1724 (0.1533)		0.1758 (0.1529)	
Scenario 4	−0.1506 (0.1447)		−0.1505 (0.1330)		−0.1458 (0.1332)	
France	−0.3585 (0.0942)	*	−0.3584 (0.0946)	*	−0.3544 (0.0943)	*
$\widehat{ICC} = 0.0002$			$\hat{\rho} = 0$		$\hat{\rho} = -0.0034$	
			QIC = 2623.41		QIC = 2623.34	

Furthermore, under the Bonferroni correction the adjusted significance level only scenario 2 had a significant effect on the odds of good pear quality with 1.6835 times (67.35% higher than) scenario 1. On the other hand, scenario 3 and 4 did not show a statistically significant effect on having a good pear quality, with the odds of 1.1880 and 0.8602 times the first

climate scenario respectively.

3.5 The effect of climate condition on pears over time

The fixed effect of model (3) was described in Table 6. It explained that along the weeks of the study period, climate and species gave a highly significant on the growth of pears at the 5% level of significance. The parameter estimates of the fixed effects were shown in

Table 6: Type 3 test of fixed effects

Effect	<i>F</i> -value	<i>Pr</i> > <i>F</i>
Species	70.86	<0.0001
Climate	0.24	0.8681
Time	30676.9	<0.0001
Time · Climate	73.35	<0.0001
Time · Species	271.01	<0.0001

Table 7. Under Bonferroni correction the adjusted significance level only scenario 2 showed a significantly growth rate compared to scenario 1, with an increase of 0.0515 cm per week. Notably, scenario 4 grew significantly slower by 0.0148 cm per week, but in the opposite direction to the hypothesis. Compared to the Conference variety, Doyenné grew slower by 0.0532 cm every week.

Table 7: Parameter estimates for fixed effect longitudinal Gaussian outcome

Variable	Est.	Std. Err	<i>t</i> -value	<i>Pr</i> > $ t $	Lower	Upper
Intercept	2.5056	0.0504	49.74	<0.0001	2.3453	2.6659
Doyenné	−0.3815	0.0453	−8.42	<0.0001	−0.4703	−0.2926
Scenario 2	−0.0223	0.0649	−0.34	0.7310	−0.1497	0.1050
Scenario 3	0.0318	0.0633	0.50	0.6152	−0.0923	0.1560
Scenario 4	0.0091	0.0634	0.14	0.8863	−0.1153	0.1334
Time	0.2996	0.0036	83.79	<0.0001	0.2926	0.3067
Time · Scenario 2	0.0515	0.0047	11.02	<0.0001	0.0423	0.0607
Time · Scenario 3	0.0048	0.0045	1.07	0.2850	−0.0039	0.0136
Time · Scenario 4	−0.0148	0.0045	−3.28	<0.0001	−0.02359	−0.00593
Time · Doyenné	−0.0532	0.0032	−16.46	<0.0001	−0.0596	−0.0469

Table 8 described the covariance estimates for the random effects from LMM for longitudinal (3), where the entries for the matrix D were given by

$$D = \begin{bmatrix} 0.0419 & -0.0026 \\ -0.0026 & 0.0004 \end{bmatrix}.$$

As described in Table 8, the variance of the random intercept at the pears level was 0.0419, indicating considerable variability in the starting size among pears within the same tree. The variance of the random slope at the pear level was small 0.0004, indicating the growth rates of pears within the same tree were relatively similar to one another.

Furthermore, the negative covariance parameter for random intercept and random slope for pears implied that pears starting with a larger size tended to grow slower, while pears starting with a smaller size tended to grow faster. The correlation between the random intercept and random slope was -0.6351 . The variance of the random intercept at the tree

Table 8: Covariance parameter estimates.

Cov. Param.	Parameter	Subject	Estimate
Intercept	σ_{tree}^2	tree	0.0015
UN(1,1)	$b_{0i(j)}$	Pear (tree)	0.0419
UN(2,1)	$Cov(b_{0i(j)}, b_{1i(j)})$	Pear (tree)	-0.0026
UN(2,2)	$b_{1i(j)}$	Pear (tree)	0.0004
Residual	σ_{res}^2		0.0525

level was considerably small by 0.0015, indicating a small variation of pear size between trees at the start of study period. The residual variance explained the remaining unexplained variability in pear size within each pear over time, after accounting for all fixed and random effects in the model.

3.6 Soil condition on the effect of climate on pears

3.6.1 Explore the soil condition features by Principal Components Analysis

Figure 4a showed the scree plot of the first ten principal components, together with the cumulative explained variance. The first three components explained the largest proportions of variance (19.5%, 14.7%, and 14.5%), after which the contribution gradually decreased, indicating that variability was spread across multiple dimensions. The cumulative curve showed that approximately 80% of the total variance was captured by the first seven components, suggesting that a reduced representation using around 7 principal components would retain most of the information. The scree plot indicated that the dataset did not exhibit a strong low-dimensional structure, and that important information was spread across several principal components.

Figure 4b illustrated the PCA biplot based on the first two principal components. However, no clear clustering or separation of the four climate scenarios was observed on the PC1–PC2 plane. Observations from different scenarios substantially overlapped and appeared to be widely dispersed. Furthermore, the loading vectors did not indicate a really clear directional

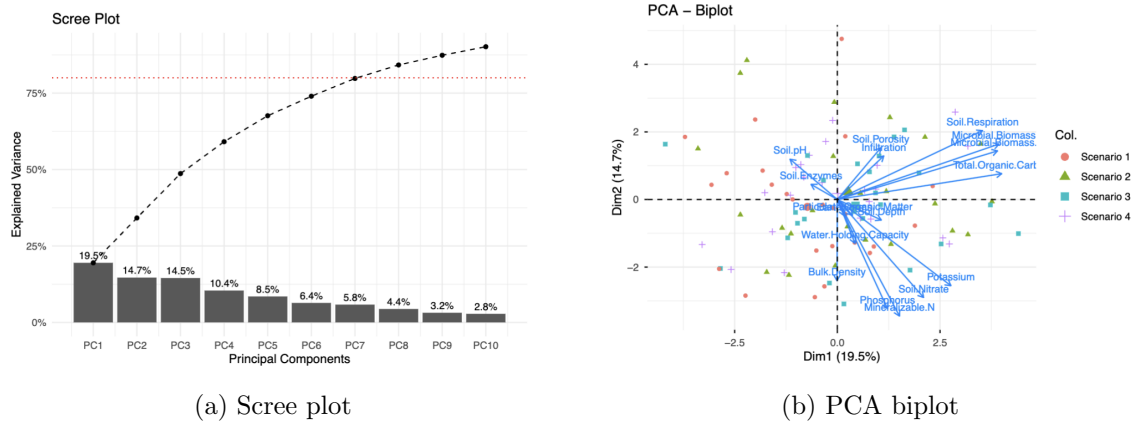


Figure 4: Visualization of PCA by scree plot and biplot.

separation, as projections of observations onto individual variable directions did not reveal consistent patterns across climate groups. This suggested that the soil features, as represented by the first two principal components, did not provide strong evidence of differences between the climate scenarios.

The PCA results indicated that based on the variation captured by PC1 and PC2, there was no strong evidence that soil conditions differ substantially across climate scenarios. This implied that if differences exist, they were likely subtle or distributed across higher-dimensional structures not captured by the first two principal components.

3.6.2 Investigate effect of climate on the pear production by Linear Discriminant Analysis

Figure 5a showed the proportion of between-class variance explained by each linear discriminant. The first linear discriminant (LD1) explains approximately 59.8% of the variance, followed by LD2 (23%) and LD3 (17.1%). This indicated that the majority of the separation between climate scenarios was captured by LD1, with additional but smaller contributions from higher-order discriminants, suggesting that group differences were primarily driven by a single dominant direction.

Figure 5b showed the LDA scatter plot based on the first two linear discriminants. Compared to PCA, a clearer separation between climate scenarios was observed, particularly along LD1. Scenario 4 tends to be located on the negative side of LD1, while Scenarios 2 and 3 are mainly positioned on the positive side, with Scenario 1 lying in between.

However, some overlap between groups remained, as indicated by the confidence ellipses, suggesting that the separation was not complete. Overall, the results indicated that soil features provide some discriminatory power between climate scenarios, although the differ-

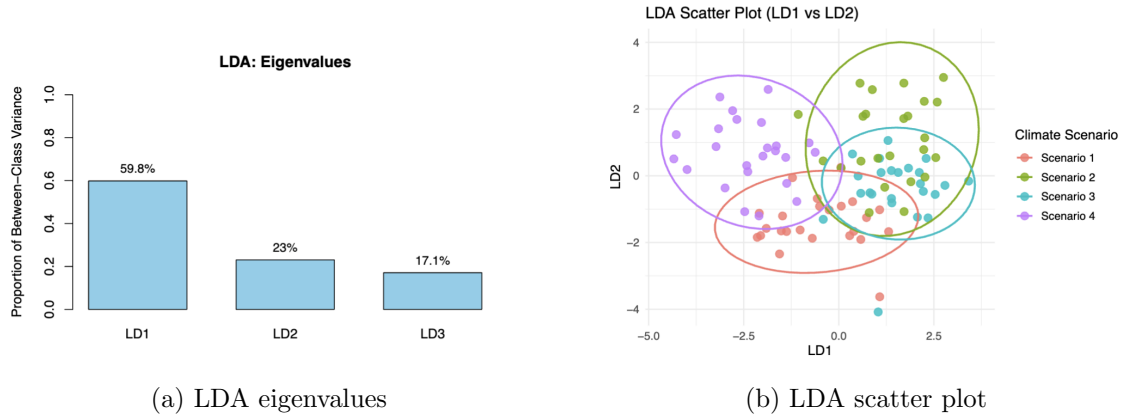


Figure 5: Eigenvalues and scatter plot of LDA, which the circles on the scatter plot showed clusters.

ences were not perfectly distinct.

Figure 6 shows the contributions of individual variables to the first linear discriminant (LD1). Variables such as bulk density, soil porosity, infiltration, and total organic carbon

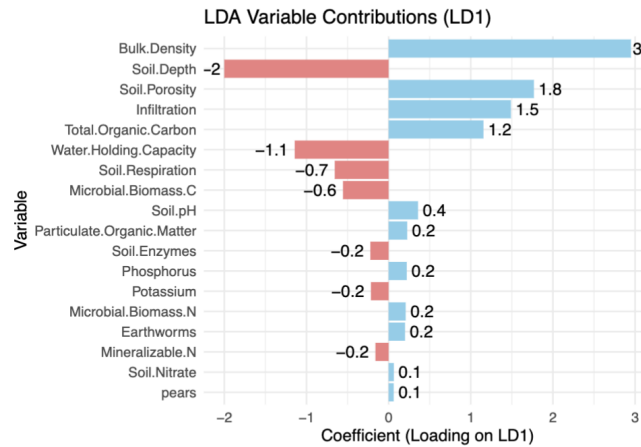


Figure 6: LDA variable contributions on LD1

had strong positive loadings, while soil depth and water holding capacity contributed in the opposite direction. This indicates that the separation along LD1 was primarily driven by contrasts between these soil characteristics.

In contrast, pear production exhibits a negligible loading, suggesting that it contributed minimally to the discrimination between climate scenarios. Overall, the results indicated that group separation was mainly driven by differences in soil conditions rather than direct

differences in pear yield.

Overall, the results suggested that soil conditions might vary slightly across climate scenarios, as indicated by the separation observed in LDA. However, these differences were not dominant in the overall variability of the data, as PCA did not show clear separation. In contrast, pear production showed negligible contribution to the discrimination, indicating that it did not differ substantially across climate scenarios. Hence, there was not enough evidence from this multivariate analysis to conclude that climate scenarios have a meaningful impact on pear production, and any potential effect was likely weak or not captured by the soil features considered.

4 Discussion

Figure 1a illustrated that the distribution of the continuous quality index was similar across the climate scenarios and locations of the Ecotrons with a wide overlapping interquartile ranges. The IQR within each group shown in Table S1 consistently ranged between 24 and 42. Additionally, the mean difference between climate scenarios, species, and locations, showed no significant difference. However, the initial LMM included the interaction between climate-location, climate-species, and species-location, but non of these were significant to estimate the pear quality index. Hence, in the final model (1) we did not include any interaction effect.

The unstructured structure was explored for LMM model, however, a linear mixed model with a single random intercept implied a compound symmetry covariance structure [5]. Additionally, they had a really similar results to each other. Hence, we reported only the compound symmetry structure in the table.

For further comparison the GEE for Gaussian outcome was provided in Table 3, where we found similar conclusion between LMM and GEE. GEE had an advantage where we were “allowed” to misspecified the working correlation. We could see whether the model was misspecified or not by comparing the standard error (SE) of model based and empirical results, which in our case they were similar to each other [5]. Since the models had similar conclusion and the SE of LMM and GEE was close to each other, indicated that our covariance structure was not misspecified. Hence, we interpreted our LMM results instead of GEE.

The binary outcome illustrated in Figure 1b gave a more visual variability than the Gaussian outcome in Figure 1a. The same interaction effects were included in the model, and only climate-location was significant. However, the only significant was France under scenario 4 but it was really close to the borderline of the significance level. Additionally, the LRT of no interaction model (2) was a slightly lower at 8305.14 compared to the model with

climate-location interaction with 8319.69. Hence, for the easiness of interpretation, we kept the model as parsimony as possible which no interaction involved.

As observed in Table 5, GEE with exchangeable and independence working correlation were also fitted for conclusion comparison purposes. With the same reason as the Gaussian outcome, the GLMM was selected as the primary model for answering the research question.

In longitudinal dataset, Figure S2 illustrated that there were some dropout observations during the 20 weeks of study period. We compared two different longitudinal data, where we kept the dropout and we removed the dropout. The results for each case was shown in Table 7 and S3, respectively. From the results, we observed similar conclusion for both cases. The only difference was the standard error in Table 7 that slightly larger than the second case. Although we did not remove the pears with dropout, the result was not bias because of full-likelihood approach [4]. By this reason, we considered to use the LMM instead of GEE to address the longitudinal research question.

From the three of our statistical models, there were several common variables that had significant effect to the pear quality (size), which were pear varieties and climate scenario. Conference variety showed a better quality and grew faster compared to Doyenné variety. Based on model (1) and (2), Doyenné was estimated to have a lower quality compared to Conference variety as the reference group. Moreover, the longitudinal model (3) showed in weekly that Doyenné tended to grow slower than Conference.

Regarding the climate scenarios, scenario 2 was the most consistently beneficial intervention across all three models, showing significantly higher pear quality, higher odds of good quality, and a faster growth rate compared to Scenario 1. On the other hand, scenario 3 showed a significant positive effect on pear quality in the first model (1), but not in the binary outcome model or longitudinal model. The last intervention showed a significant slower growth rate compared to scenario 1 in the longitudinal model, and did not give any significant effect in model (1) and (2).

PCA did not reveal strong clustering among climate scenarios, suggesting that climate-related variation was not dominant in the overall soil variability (Figure 4b). However, LDA achieved clearer separation between groups (Figure 5b), indicating that subtle differences may still exist when group information is incorporated. This difference likely reflects the different objectives of the two methods, as PCA focuses on overall variance whereas LDA maximizes between-group discrimination. Nevertheless, the relatively low explained variance and overlap between groups suggest that further analyses are needed before drawing strong conclusions about the influence of climate conditions on pear production through soil characteristics.

5 Conclusion

To conclude, we observed that Doyenné was estimated to have a lower quality by 6.35 and the odds of having a good quality was only 0.6 times the Conference variety. Notably, Doyenné was estimated to have a slower growth rate compared to Conference variety. On top of that, after adjusted the significance level by Bonferroni correction, climate scenario 2 was the only intervention that was estimated by all three statistical models that had a significant effect to the quality index, good quality, and the size over the week. Nevertheless, LDA suggested a better separation between climate scenarios compared to the PCA.

References

- [1] Stefano Musacchi, Ignasi Iglesias, and Davide Neri. Training systems and sustainable orchard management for european pear (*pyrus communis* l.) in the mediterranean area: A review. *Agronomy*, 11(9):1765, 2021.
- [2] Inne Vanderkelen, Jakob Zscheischler, and et al. A novel method for assessing climate change impacts in ecotron experiments. *International journal of biometeorology*, 64(10):1709–1727, 2020.
- [3] Jessica M Waite, Christopher Gottschalk, Lauri A Reinhold, Nahla V Bassil, Gayle M Volk, Joseph D Postman, Rachel B Elkins, and Richard L Bell. Vulnerability of pear (*pyrus*) genetic resources in the us. *Genetic Resources and Crop Evolution*, 72(1):815–843, 2025.
- [4] Geert Molenberghs and Geert Verbeke. *Linear mixed models for longitudinal data*. Springer, 2000.
- [5] Geert Molenberghs and Geert Verbeke. *Models for discrete longitudinal data*. Springer, 2005.
- [6] Wei Pan. Akaike’s information criterion in generalized estimating equations. *Biometrics*, 57(1):120–125, 2001.
- [7] Brian Everitt, George Casella, Stephen Fienberg, and Ingram Olkin. *An R and S-PLUS companion to multivariate analysis*. Springer, 2005.
- [8] Wolfgang Härdle and Léopold Simar. *Applied multivariate statistical analysis*. Springer, 2015.

Appendix

Table S1: Mean, quartiles (Q_1 , Q_3), and interquartile range (IQR) quality index across climate scenarios.

Climate	Location	Species	Mean	Q_1	Q_3	IQR
Scenario 1	BE	Conference	55.07	39.00	73.00	34.00
	BE	Doyenne	49.93	28.50	70.75	42.25
	FR	Conference	51.17	31.00	69.50	38.50
	FR	Doyenne	46.33	31.00	60.00	29.00
Scenario 2	BE	Conference	63.41	46.00	80.00	34.00
	BE	Doyenne	59.91	47.00	76.00	29.00
	FR	Conference	61.89	46.00	81.00	35.00
	FR	Doyenne	58.32	43.50	75.50	32.00
Scenario 3	BE	Conference	63.18	46.00	80.00	34.00
	BE	Doyenne	56.04	38.50	73.00	34.50
	FR	Conference	57.38	42.75	77.00	34.25
	FR	Doyenne	46.94	31.75	63.00	31.25
Scenario 4	BE	Conference	60.38	46.25	77.75	31.50
	BE	Doyenne	48.47	30.50	66.25	35.75
	FR	Conference	50.22	37.00	65.00	28.00
	FR	Doyenne	43.52	31.00	55.00	24.00

Table S2: Contrasts of climate scenarios.

Scenario	Estimate	Std. Err	t -value	$Pr > t $
2 - 1	9.9964	1.6390	6.10	<0.0001
3 - 1	5.5843	1.7082	3.27	0.0011
4 - 1	0.4953	1.8087	0.27	0.7842

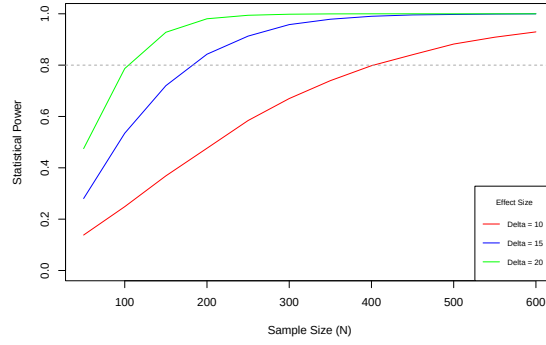


Figure S1: Sample size simulation by different effect size δ

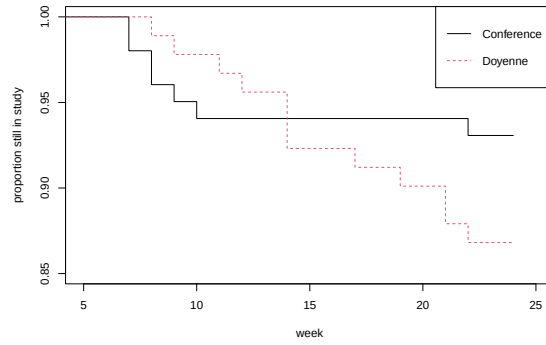


Figure S2: Representation of dropout pears left in the study until week 24.

Table S3: Parameter estimates for fixed effect longitudinal Gaussian outcome with removing the dropout.

Variable	Est.	Std. Err	<i>t</i> -value	<i>Pr</i> > <i>t</i>	Lower	Upper
Intercept	2.5093	0.0465	53.98	<0.0001	2.3614	2.6573
Doyenné	−0.3963	0.0418	−9.48	<0.0001	−0.4783	−0.3143
Scenario 2	−0.0305	0.0605	−0.50	0.6144	−0.1491	0.0881
Scenario 3	0.0414	0.0584	0.71	0.4783	−0.0731	0.1558
Scenario 4	0.0196	0.0583	0.34	0.7367	−0.0946	0.1338
Time	0.2992	0.0036	83.44	<0.0001	0.2922	0.3063
Time · Scenario 2	0.0532	0.0047	11.30	<0.0001	0.0439	0.0624
Time · Scenario 3	0.0046	0.0045	1.03	0.3011	−0.0042	0.01344
Time · Scenario 4	−0.0154	0.0045	−3.44	0.0006	−0.0242	−0.0066
Time · Doyenné	−0.0511	0.0032	−15.82	<0.0001	−0.0574	−0.0448

```

libname datasets '/home/u64375008/';
PROC IMPORT DATAFILE= '/home/u64375008/G12.outcome.data.csv'
            OUT= datasets.hierarchy
            DBMS=CSV REPLACE;
            GETNAMES=YES;
            GUESSINGROWS=MAX;
RUN;
/*LMM*/
proc mixed data=datasets.hierarchy method=reml;
    class climate(ref = "Scenario 1") species(ref = "Conference")
        location(ref = "BE") ecotr_id tree_id;
    model quality_idx = species climate location/ solution cl ddfm=satterthwaite
        outpm=predmean outp=pred;
    random intercept / subject=tree_id(ecotr_id) type=cs v g;
    estimate 'Scenario 2 vs Scenario 1' climate 1 0 0 -1 / cl;
    estimate 'Scenario 3 vs Scenario 1' climate 0 1 0 -1 / cl;
    estimate 'Scenario 4 vs Scenario 1' climate 0 0 1 -1 / cl;
run;
PROC IMPORT DATAFILE= '/home/u64375008/longitudinal_data.csv'
            OUT= datasets.longdata
            DBMS=CSV REPLACE;
            GETNAMES=YES;
            GUESSINGROWS=MAX;
RUN;
data datasets.longdata_clean;
    set datasets.longdata;
    quality_numeric = input(quality, ?? best32.);
    time_numeric = input(time, ?? best32.);
    drop quality time;
    rename quality_numeric = quality time_numeric = time;
run;
proc mixed data=datasets.longdata_clean;
    title 'No Removal';
    class climate(ref = "Scenario 1") species(ref = "Conference")
        tree_id pear_id;
    model quality = species climate time climate*time species*time/
        solution cl outp=longresults outpm=longpredmean;
    random intercept / subject=tree_id;
    random intercept time / subject=pear_id(tree_id) type=un;
    contrast 'Any climate scenario vs climate scenario 1'

```

```

        climate*time 1 0 0 -1,
        climate*time 0 1 0 -1,
        climate*time 0 0 1 -1;
    estimate 'climate scenario 2 - climate scenario 1' climate*time
    1 0 0 -1;
    estimate 'climate scenario 3 - climate scenario 1' climate*time
    0 1 0 -1;
    estimate 'climate scenario 4 - climate scenario 1' climate*time
    0 0 1 -1;
run;
PROC IMPORT DATAFILE= '/home/u64375008/quality_binary.csv'
        OUT= datasets.binary
        DBMS=CSV REPLACE;
        GETNAMES=YES;
        GUESSINGROWS=MAX;
RUN;
proc genmod data=datasets.binary;
    class climate(ref = "Scenario 1") species(ref = "Conference") location(ref = "BE") tree_id
    model quality_b(ref = "0") = species climate location / dist= binomial type3 link=logit;
    repeated subject = tree_id / type=exch covb corrw modelse;
    estimate 'Scenario 2 vs Scenario 1' climate 1 0 0 -1;
    estimate 'Scenario 3 vs Scenario 1' climate 0 1 0 -1;
    estimate 'Scenario 4 vs Scenario 1' climate 0 0 1 -1;
run;
proc glimmix data=datasets.binary method=rspl;
    class climate(ref = "Scenario 1") species(ref = "Conference") location(ref = "BE") ecotr_id
    model quality_b(ref = "0") = species climate location / dist=binary link=logit solution cl;
    random intercept / subject=tree_id(ecotr_id) type=cs v g;
    estimate 'Scenario 2 vs Scenario 1' climate 1 0 0 -1 / cl;
    estimate 'Scenario 3 vs Scenario 1' climate 0 1 0 -1 / cl;
    estimate 'Scenario 4 vs Scenario 1' climate 0 0 1 -1 / cl;
run;

library(ggplot2)
library(MASS)
library(tidyverse)
library(scales)
library(factoextra)

data <- read.csv("Data/G12.soil.data.csv")

```

```

cols_to_remove<-c("tree_ID","location","ecotr_id","tree_id","species",
"climate")
soil_data<-select(data,-all_of(cols_to_remove))
pca_data<-prcomp(soil_data,center=TRUE,scale=TRUE)
pca_var <- pca_data$sdev^2
pca_var_ratio <- pca_var / sum(pca_var)
pca_cum_ratio <- cumsum(pca_var_ratio)
n_pc <- 10
scree_df <- data.frame(
  PC = paste0("PC", 1:n_pc),
  Index = 1:n_pc,
  Variance = pca_var_ratio[1:n_pc],
  Cumulative = pca_cum_ratio[1:n_pc])

ggplot(scree_df, aes(x = Index)) +
  geom_col(aes(y = Variance)) +
  geom_line(aes(y = Cumulative), linetype = "dashed") +
  geom_point(aes(y = Cumulative)) +
  geom_text(
    aes(y = Variance, label = paste0(round(Variance * 100, 1), "%")),
    vjust = -0.5,size = 3) +
  geom_hline(yintercept = 0.8, linetype = "dotted", color = "red") +
  scale_x_continuous(breaks = scree_df$Index,
                     labels = scree_df$PC) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Scree Plot",
       x = "Principal Components",
       y = "Explained Variance") +
  theme_minimal()

fviz_pca_biplot(pca_data, repel = FALSE,
                geom.ind = "point", col.var = "#0083FF",
                col.ind = data$climate, labelsiz = 3)

pear_count <- summarise(
  group_by(outcome_data, location, ecotr_id, tree_id),
  pears = n())
soil_with_pears <- left_join(
  data,pear_count,
  by = c("location", "ecotr_id", "tree_id"))

```

```

predictors <- select(soil_with_pears,-tree_ID,-location,
                    -ecotr_id,-tree_id,-species,-climate)
group <- as.factor(data$climate)
predictors_scaled <- scale(predictors)
lda_model <- lda(group ~., data = as.data.frame(predictors_scaled))
lda_result <- predict(lda_model)
lda_scores <- as.data.frame(lda_result$x)
lda_scores$climate <- data$climate
lda_var <- lda_model$svd^2
lda_var_ratio <- lda_var / sum(lda_var)

midpoints <- barplot(lda_var_ratio,
  names.arg = paste0("LD", 1:length(lda_var_ratio)),
  col = "skyblue", ylab = "Proportion of Between-Class Variance",
  main = "LDA: Eigenvalues", ylim = c(0, 1),width = 0.3,space = 0.5)
text(x = midpoints,y = lda_var_ratio,
  labels = paste0(round(lda_var_ratio * 100, 1), "%"),
  pos = 3,cex = 0.9)
ggplot(lda_scores, aes(x = LD1, y = LD2, color = as.factor(climate))) +
  geom_point(size = 3, alpha = 0.8) +
  stat_ellipse(aes(group = as.factor(climate)),
    type = "t",level = 0.95,linewidth = 0.8) +
  labs(title = "LDA Scatter Plot (LD1 vs LD2)",
    x = "LD1", y = "LD2", color = "Climate Scenario") +
  theme_minimal(base_size = 12)
ld1_loadings <- lda_model$scaling[, 1]
loading_df <- data.frame(Variable = names(ld1_loadings),
  LD1 = ld1_loadings)
loading_df$Color <- ifelse(loading_df$LD1 < 0, "lightcoral", "skyblue")
loading_df <- mutate(loading_df, Variable = fct_reorder(Variable, abs(LD1)))
ggplot(loading_df, aes(x = Variable, y = LD1, fill = Color)) +
  geom_col() +
  geom_text(  aes(label = round(LD1, 1)),
    hjust = ifelse(loading_df$LD1 > 0, -0.2, 1.2)) +
  scale_fill_identity() +
  coord_flip() +
  labs(title = "LDA Variable Contributions (LD1)",
    x = "Variable", y = "Coefficient (Loading on LD1)" ) +
  theme_minimal(base_size = 12)

```